

Shehzade Manzoor Khan



About me

I'm Shehzade Manzoor Khan, postgraduate from physics NISER, Bhubaneswar. With a strong foundation in physics and a fervent interest in experimental physics, I am enthusiastic about the prospect of delving into new areas of research and discovery, and I am committed to contributing meaningfully to the advancement of knowledge in these fields.

Hard Skill

- Basic Knowledge in Linux, Ubuntu, OriginLab, Solid-Works(3D printing) and XPS casa
- Intermediate Knowledge: Excel, Word and Power-Point
- Used SEM, EPMA, UV spectrometer and petrographic microscope
- Sample preparation of meteorite for SEM.
- Setting up of 6 inch telescope.

Soft skills

- Observation
- Decision making
- Communication
- Multi-tasking
- Organizational Skills
- Previous active member of NISER Astronomy Club (NAC)
- Previous member of Robotics Club of NISER (3D Printing Division)
- Previous core committee member of SAPSARA (Photography Club of NISER)

Languages

Malayalam	C2	Proficient
English	C2	● ● ● ● ●
Hindi	C1	● ● ● ● ●
French	A1	● ● ● ● ●

RESEARCH EXPERIENCE

May 2024
- October 2025

Redox-Controlled Third-Order Nonlinear Optical Response in thiazine dyes and Fabrication of Distributed Bragg Reflector NIT · Calicut

I recently conducted an experimental study on the reduction of thiazine dyes, with a focus on understanding the reaction kinetics and the change in optical properties during the redox process. The work involves preparing precise concentrations of dye and reducing agent, while performing UV-Visible spectroscopy measurements, and analyzing absorbance decay to determine reaction under different conditions. Open-aperture Z-scan measurements were performed using a Q-switched Nd:YAG laser operating at a wavelength of 532 nm with a pulse duration of 7 ns and a repetition rate of 10 Hz. The change of the nonlinear response is attributed to the disruption of the conjugated electronic structure upon reduction, which reduces electronic delocalization and optical absorption in the visible region. I also successfully fabricated and characterized a 10.5 bilayer Dielectric Bragg Reflector (DBR) optimized for quarter-wavelength ($\lambda/4$) thickness. The film thickness was fine-tuned by adjusting the spin coater speed and varying the concentration of the PVK and CA solutions. This method allowed us to optimize the DBR effectively. The refractive indices of PVK (1.683) and CA (1.475) provided sufficient contrast, resulting in a DBR with a reflectance of approximately 97.2%.



Summer 2023

Neon composition in bulk Carbonaceous Chondrites

PHYSICAL RESEARCH LABORATORY (PRL) · Ahmedabad

Completed summer internship 2023, from PRL, Ahmedabad, on 'Neon composition in bulk Carbonaceous Chondrites' under Ramakant R Mahajan. The study aimed to investigate the composition of neon (Ne) within bulk carbonaceous chondrites. The primary objective was to determine the concentration of trapped ^{20}Ne in these meteorites. The research also involved the generation of Neon 3-isotope graphs, incorporating end member components tailored to various types of carbonaceous chondrites.



Summer 2022

Analysis of Brownian Dynamics

IIT · Mandi

Completed Summer internship 2022, from IIT Mandi, on 'Analysis of Brownian Dynamics' under Prof. Harsh Soni. The project was centered around examining the dynamics of a single Brownian particle utilizing both theoretical frameworks and computer simulations. Two distinct potentials, namely the simple harmonic potential and the double well potential, were investigated to analyze the Brownian motion of a microparticle. Following our theoretical understanding, position-time graphs were generated to visualize the particle's behavior under each potential.



August 2022

National Workshop on Coronal Mass Ejections

CENTURION UNIVERSITY · Bhubaneswar

Attended National workshop for CME Kinematics 2022 at Centurion University. Images and movies derived from the SOHO telemetry were utilized in near real-time to monitor the current state of the Sun. These resources proved invaluable for detecting phenomena such as sunspots, solar flares, and coronal mass ejections (CMEs).



Summer 2021

QIQT Summer School

IISER · Kolkata

I had attended tutorials and sessions covering a wide spectrum of quantum topics, including quantum information, quantum optics, quantum computation on various platforms, quantum simulation/computation, quantum chemistry, and quantum sensing/imaging applications, my understanding of quantum physics has undergone significant enhancement. These sessions have provided me with invaluable insights into both the theoretical foundations and practical applications of quantum principles.



CERTIFICATES & GRANTS

1-2 April 2026	Poster Presentation "Redox-Controlled Third-Order Nonlinear Optical Response in Methylene Blue" at International Conference on Optics, Lasers and Photonics (OPTIX 2026), organized by the NIT Calicut Optics Student Chapter, under Nonlinear Optics and Optical Materials (DOI: 10.26643/edupub/978-93-92585-01-2).
2026	Qualified GATE PH .
2026	Graduate Record Examination (GRE): Total Score: 315
2025	TOEFL iBT: Total Score: 109/120
4-7 March 2025	Poster Presentation "Thermal Evolution of Planetesimals: A Case for Asteroid Vesta" under the theme Planetary Science and Exploration at the International Conference on "Space for Sustainability: Science, Technology, Education and Policy (S2:STEP 2025)" and the 6th Indian Planetary Science Conference (IPSC-2025), held at the IIT Roorkee, India.
26 Feb-1 March 2024	Poster Presentation at the 22nd National Space Science Symposium, Goa University, under PS3 – Solar and Planetary Sciences for the topic "Neon Composition in Bulk Carbonaceous Chondrites".
1-3 November 2023	Presented poster titled "Neon Composition Study in Bulk Carbonaceous Chondrites" at MetMESS 2023 held at Physical Research Laboratory (PRL), Ahmedabad.
4-7 March 2025	Presented poster titled "Thermal Evolution of Planetesimals: A Case for Asteroid Vesta" under the theme Planetary Science and Exploration at the International Conference on "Space for Sustainability: Science, Technology, Education and Policy (S2-STEP 2025)" and the 6th Indian Planetary Science Conference (IPSC-2025), Indian Institute of Technology Roorkee.
2020 - 2025	DISHA Scholarship from Department of Atomic Energy (DAE).
2020	Qualified KEAM, JEE Main, and National Eligibility Cum Entrance Test (NEET) .
2013-2014	Selected as "Bright Spot" in the Spot Assessment conducted annually by Vikram Sarabhai Science Foundation (VSSF).

CURRICULUM

Courses Taken	<i>Introduction to Physics-I & II, Classical Mechanics, Electronics, Mathematical Methods-I & II, Quantum Mechanics-I & II, Electromagnetism-I, Group Theory, Linear Algebra, Special Theory of Relativity, Nuclear And Particle Physics, Statistical mechanics, Theory of Computation</i>
Msc. Thesis	<i>Modeled the thermal evolution and internal differentiation of early-forming planetesimals, with a focus on asteroid Vesta, using a 1D spherical heat-conduction framework. Initial simulations were performed using the Crank-Nicolson method. The model was further improved using the Method of Lines (MoL). This approach enabled more accurate and flexible treatment of the thermal evolution across all stages</i>

INTERESTS AND ACTIVITIES

- Completed 5K Republic Day marathon, NISER 2024.
- First, second, and second prize respectively for Shotput, Discus throw, and Javelin throw in Tvisha.
- 2 times gold medallist in shotput IISM.
- Participated in SSU, NLU, and Chiasma (AIIMS, Bbsr) volleyball tournament.
- Elected Captain of volleyball team of NISER
- Won third place in Centurion volleyball tournament.
- Fourth grade certificate for piano from Trinity College London.
- Got Second place for State CBSE Sahodaya 2018 for Drama.

EDUCATION BACKGROUND

Degree	Integrated MSc.	
	5 YEARS(DECEMBER 2020 - JULY 2025) · National Institute of Science Education and Research, Bhubaneswar 	

PROGRAMMING

C, C++	
LaTeX	
python	
Shell scripting	
Labview	